# EGB 387 – Engineering Economy & Planning
# Project Assignment 2026

## *ENOGGERA RESERVOIR HYDRO PROJECT*

*Submission date 11th September (23.59) 2026*

## ENOGGERA RESERVOIR HYDRO PROJECT

### Preamble:

For the last several years, homeowners have been encouraged to install solar panels on the roofs of residential properties. The encouragement has taken the form of substantial payments made by the Government for any electricity generated beyond the needs of the household. The installation cost of the panels has also been subsidised.

As a consequence, owners have installed numbers of panels well beyond that required to provide hot water – the only practical form of storing this energy. The high cost and physical size of batteries needed to store this surplus energy for use at night and at times when the solar generation is seriously degraded, has prevented this as an option. Rather, the surplus is fed back into the mains and any shortfall in energy generated by a household at other times is made up by drawing on the normal mains supply.

The result is a flow of current into the mains at times of strong sunlight causing serious problems for the supply Authorities and a heavy demand on supply at times of low solar activity. The effect is random and does little to reduce the level of normal generation.

### The Project:

To accommodate these surges, a project has been developed to use the surplus to pump water to an elevated dam for later release through hydro-electric generators to return energy to the system. The flexibility of this system permits matching of the supply to times of peak demand.

The proposed location of the scheme is at the Enoggera Reservoir. This site offers the advantages of close proximity to a large source of solar energy supply.

The proposal requires the construction of several civil engineering structures together with ancillary mechanical and electrical systems:

1. A dam in the foothills immediately westwards of the Reservoir;

2. A small Power Station;

3. A Pumping Station;

4. A steel penstock from the dam to the combined Pump and Power Station;

5. A transmission line connection to the suburban electrical system adjacent to the site and

6. Ancillary tasks, including dredging in the ponded area of the Reservoir and road works.

### Scope of the assignment:

You are a recent graduate working for a medium sized construction company which is tendering for the construction of the facility. You have been set the task of assisting the company estimators by developing the construction procedures to be used for the three major areas of work – the dam, the power/pump station and the major pipeline.

The Client has decided that the small access bridge to the Pump and Power Station depends on the construction methods adopted and the degree of site assembly envisaged by the Electrical and Mechanical Contractors since this will affect the loading. For this reason, the bridge is to be bid as a Design and Construct component of the overall tender. Once the methods are decided, you are to select the equipment needed and identify such temporary works as are required to implement these methods. To do this you will need to undertake the conceptual design of these temporary works.

Your role in the bridge D & C component is to investigate alternatives and to select suitable construction methods for your chosen design. You are not required to undertake a full design of the bridge – this will be done by others in the company.

You will also be required to develop a preliminary construction program that ensures completion in the allocated time. From this program you will be required to assess the supervision demands of the work, the working hours and the time that the major plant will be onsite. In addition, you need to deal with stakeholders, in particular the Aboriginal and Torres Strait Islands, the impact of the project on them.

Three companies have been nominated by the Client to provide the electrical, mechanical and dredging functions. The coordination of these activities is part of your brief as the Principal Contractor.

You are not required to cost your methods in this Unit – it is intended that this will be done in a later Estimating Unit. Since you do not have recourse to costing as a final determinate of your choices, the decisions are to be judged on the appropriateness of the methods chosen to cope with the physical conditions inherent in the work. In this regard you will be given guidance during lectures and tutorials.

**Assignment details:**
**Weighting:** **2**5%
**Assessment Type:** Group
**Group / Individual:** Group (15%), individually (10%), Group and individual submission

**Unit Learning outcomes measured**

- ULO 1 Understand different aspects of project planning, life cycle, culture and quality.
- ULO 2 Demonstrate the importance of connectivity of project time, cost, risk and quality

**What you need to do**

A briefing document will be provided detailing the project; you are required to work in a group of 4 or 5 **and finalise it by week 2 latest**. Please discuss with your classmates and form a group of 4–5 members before joining a Canvas self sign-up. Once your group is confirmed, all members should join the same Canvas group; please do not join any group randomly without prior agreement from the group members. You should work on this project over a period of 7 weeks to produce a project executive plan for the Enoggera Hydro Dam project; it is expected you will report progress on a weekly basis as part of the submission; your contribution will be assessed via your engagement with the group.

**The submission :** It is an individual and group submissions comprising the followings:

**Part 1. Internal Briefing Document** (This is a group submission)

You are to prepare a briefing document suited for internal company use in preparing the tender. The document will comprise the sections set out below:

1. The supervision team that you propose for the project with the duration of each senior person onsite nominated. The availability of each is to be addressed and any deficiencies resolved.

2. A preliminary construction program that indicates the timing and duration of all significant phases of the work and the interrelationship of tasks.

3. A plan of the temporary roads, work areas and the deployment of site buildings that you propose.

4. A detailed description of the methods you would adopt for the four areas of construction listed below:

   - Construction of the rolled concrete dam including obtaining the materials.
   - Construction of the Power and Pump Station.
   - Construction of the concrete access bridge.

- Construction of the pipeline.

The text of your submission is to be supported by sketches. These sketches may be freehand if desired but must be clear and orderly. Submissions without adequate sketches will be deemed incomplete and will be rejected.

**Part 2. Reflective Report** (This is an individual submission)

On completion and submission of the above document, you are each required to prepare a written Report reflecting generally on the **processes** adopted by you and your Group in decision making.

In addition to the generic approach, each Report should specifically address the way the decisions were made in one of the areas below:

1. The composition of the supervisory team
2. The provision of materials for the dam construction.
3. The method you recommended for the underwater work associated with the construction of the Power and Pump Station
4. The reasoning behind your adoption of temporary works.

Each member of the group is to select one of the above topics for an expanded discussion on the thought process used in arriving at the decisions for that task. Each report is to be certified as being the individual work of that member by a signed statement to that effect.

In assessing the quality of this section, evidence of a clear line of reasoning in arriving at your chosen methods will be the principal criteria. It will not matter that you did not adopt a method that could be expected of a seasoned constructor – this will come with experience. An honest appraisal of the process you used will serve to reinforce the lessons you have learned from the assignment. Further assistance on the structure of this Report will be given in tutorials.

**Degree of effort:** There will be a high level of integration of the lectures and tutorials with the project. The teaching is heavily experienced based and sporadic attendance at either lectures or tutorials will result in an inadequate understanding.

**Additional Information**

**Timing**

The project is to be awarded at the end of March, 2027 and is due for completion by 31st May, 2029. All environmental approvals have been secured.

Working hours are limited to 10 hours per day and are to be arranged to ensure no noise nuisance occurs after 6.00pm. The working week is restricted to six days per week except in emergency situations.

**Your Company**

You are employed by a medium sized civil construction company located at Sumner Park in Brisbane. Your company has experience in earthworks and in complex structures, particularly concrete structures. You undertake as much of the work as practicable by directly employing labour, using sub-contractors only for building trade functions associated with the Pump and Power Station and curtain grouting to dam cut off.

**Assessment of the company's capacity to undertake the work**

1. The outlay on capital expenditure for equipment is small
2. Other company commitments are not excessive through period
3. Availability of suitable company personnel.

| | |
|---|---|
| Project Manager | completes current project in November 2026 |
| Project Engineers (2 required) | only 1 available October 2026. |
| Earthworks Foreman | available end January 2027 |
| Concreting Foreman | available end October 2026 |
| General Foreman | can be made available in October 2026 |
| Design Office capacity | adequate after October 2026 |

4. Availability of Company equipment:

| PLANT ITEM | DESCRIPTION | INSPECTION, TRANSPORT ASSEMBLE | SOURCE | REMARKS |
|---|---|---|---|---|
| Dozers | Two Caterpillar D8, equipped with rippers | 5 days | Owned | Own low loader |
| Elevating Scraper | Caterpillar 613 | 3 days | Owned | Own low loader |
| Scraper | Caterpillar 621 | 3 days | Owned | Own low loader |
| Excavator | Caterpillar 320, hydraulic, track mounted | 5 days | Owned | Own low loader |
| Grader | Caterpillar 120G | 3 days | Owned | Self travelling |
| Wheeled loader | Caterpillar 930 | 5 days | Owned | Own low loader |
| Backhoe/ loader | Caterpillar 426 | 3 days | Owned | Own low loader |
| Compactor | Caterpillar 815 | 5 days | Owned | Own low loader |
| Vibrating roller | Dynapac CA 25, smooth drum | 10 days | Owned | Own low loader |
| Tip trucks | Two 8m3 tandem drive | 5 days | Owned | Self travelling |
| Pile frame | Steel lattice construction 2m x 2m plan area<br>Extends to 30m in 4m increments<br>Mounted in steel base 6mx4m | 10 days | Owned | Own transport<br>Wt 8t |
| Mobile cranes | Two units, lattice jib 30m<br>Capacity 16t @ 20m | 3 days | Owned | Self travelling |
| Compressors | Two units 600 cfm, diesel powered | 3 days | Owned | Wt 9t total |
| Yard crane | Capacity 8t<br>Diesel powered, 1 owned | 2 days | Owned | Self travelling |
| Pile driving hammer | Single acting<br>Ram wt 6t | 15 days | Owned | Road 7t |
| Pile driving hammer | Double acting | 15 days | Owned | Road 5t |
| Sheetpile crib | Steel, timber wales | 10 days | Owned | Road 3t |
| Air receiver | Wheel mounted, 4 outlet manifold | 3 days | Owned | Road 2t |
| Pile helmet | Suit 400dia steel tube piles | 30 days | Owned | Road 1t |
| Concreting plant | Compressors, placing buckets, vibrators, finishing equipment | 10 days | Owned | Say 20t |
| Dewatering pumps | 200 dia centrifugal, diesel motor<br>2 available | 5 days | Owned | Road |
| Welding equipment | Diesel powered welder, oxy equipment | 3 days | Owned | Road |

| | | | | |
|---|---|---|---|---|
| Paving breakers etc. | Air operated, incl. steels and hoses | 2 days | Owned | Road |
| Steel dumb barges | Two units, 13m x 6m x 1.5 depth, incl, bridging beams | 10 days | Owned | Road |
| Workboats | Three units, 6m x 2m x 1m each powered by 50 hp outboard motor | 10 days | Owned | Road |
| Minor plant and consumables | Miscellaneous | On demand | Buy locally | Road |

5. Availability of Company owned salvaged materials

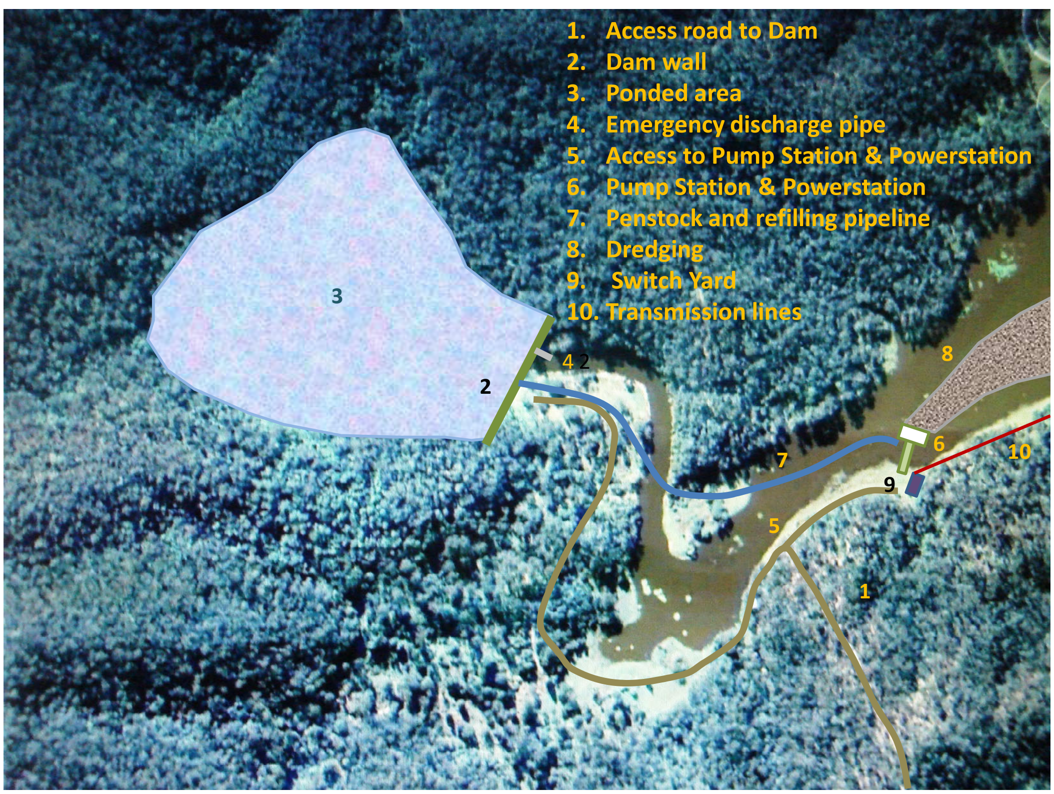
| **PLANT ITEM** | **DESCRIPTION** | **INSPECTION, TRANSPORT AND ASSEMBLY TIME** | **SOURCE** | **REMARKS** |
|---|---|---|---|---|
| Steel beams (salvaged) | 610 x 230 UB 125kg/m 330 M Some light corrosion | Available July | Gladstone | Transport by road Wt 42t |
| Steel sheet piling | BHP 50 lb section 1900 m random lengths, some light corrosion | Stacked | Emerald | Transport by road Wt 145t |
| Steel tube (salvaged) | 400 dia x 10mm wall, 100 M | Immediate | Own yard | Transport by road Wt 12t |
| Sheetpile corners | Fabricated in 12 m lengths 4 off | Stacked | Emerald | Transport by road Wt 5.5t |
| Second hand steel sections | Misc beams, columns ex marine falsework, limited corrosion | 20 days | Own yard | Transport by road Say 30t |

6. Availability of externally hired plant:

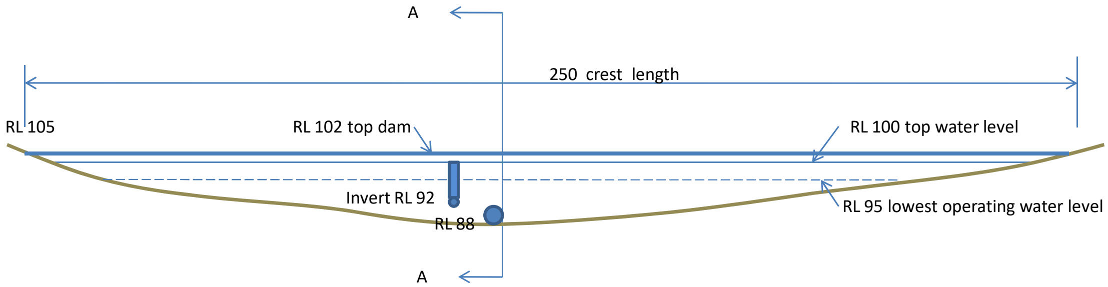
| **PLANT ITEM** | **DESCRIPTION** | **INSPECTION, TRANSPORT AND ASSEMBLY TIME** | **SOURCE** | **REMARKS** |
|---|---|---|---|---|
| Excavator | Hitachi ZX 120 | 7 days | Brisbane | Road |
| Concrete batch plant | 5m3 complete with all ancillary equipment | 30 days | Gladstone | Road |
| Agitator trucks | Two x 5m3 | 20 days | Brisbane | Road |
| Dump trucks | Four Volvo 20t | 10 days | Brisbane | Road |
| Tip trucks | Two 8m3 tandem drive | 5 days | Brisbane | Self travelling |
| Tracked crane | 150t Hitachi | 30 days | Brisbane | Road low loader |
| Lifting gear | Misc purpose made slings | 40 days | External purchase | Road 5t |
| TIG Welders | Diesel powered | 19 days | Brisbane | Road |
| Mobile crane | Capacity 25t. telescopic jib | Self travelling | Brisbane | Road |
| Yard crane | Capacity 8t Diesel powered | Self travelling | Brisbane | Long term hire |
| Scaffolding | Steel tube, metal planks | On demand | Brisbane | Long term hire |

**Conceptual Drawings**


LOCATION OF RESERVOIR



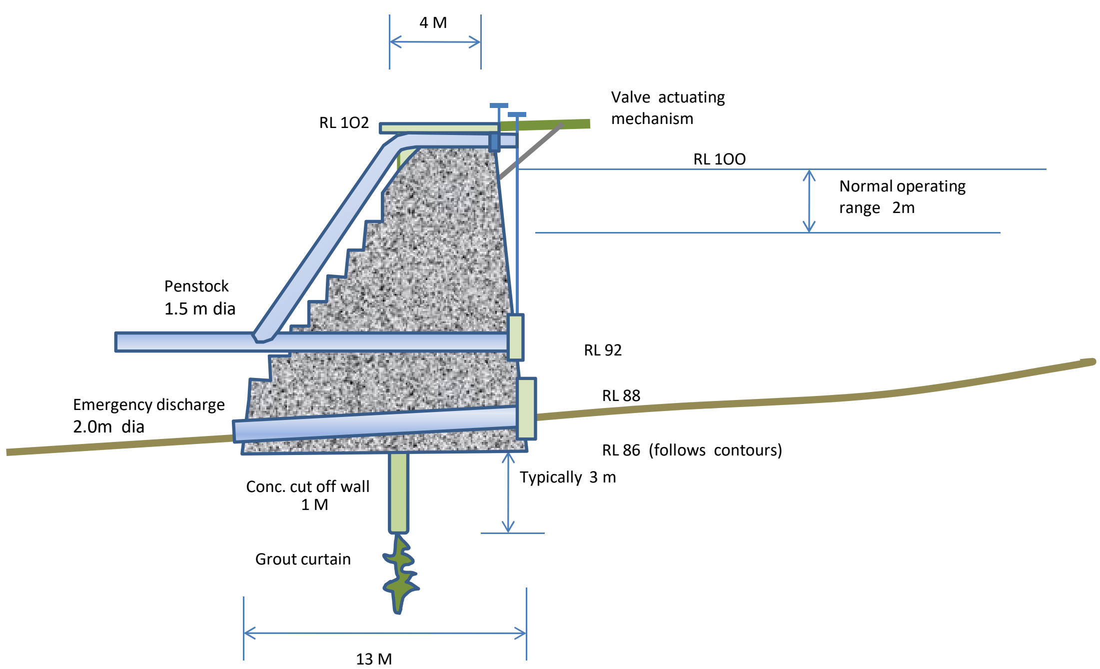

SCOPE AND SEQUENCE OF OPERATIONS




**ELEVATION OF DAM (EAST FACE)**

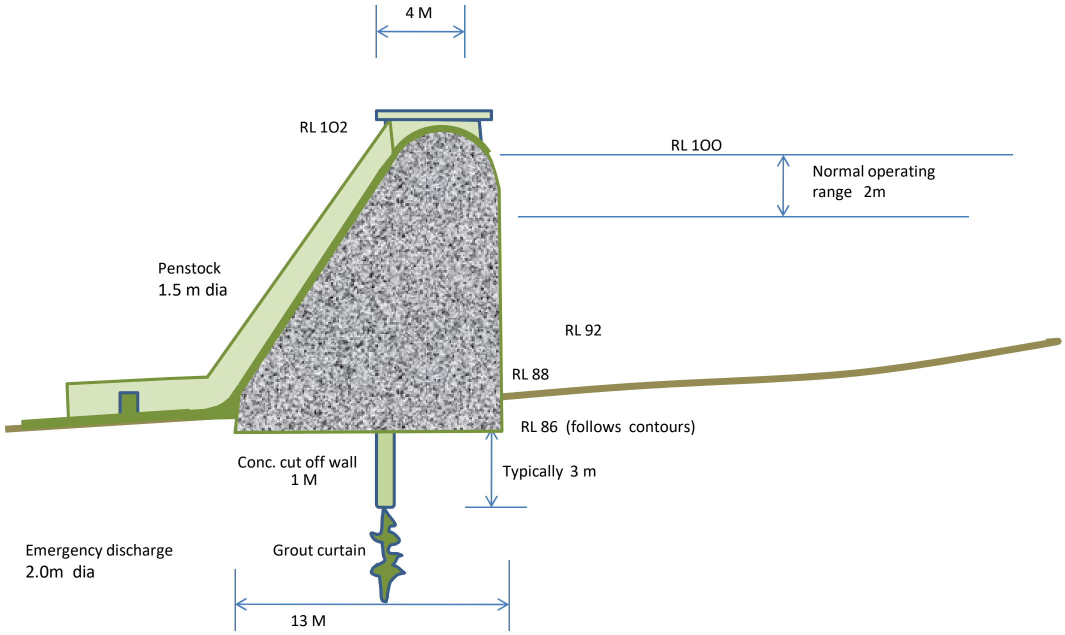

**SECTION A - A**

Not to scale




**ROLLED COMPACTED DAM**

Not to scale




**ROLLED COMPACTED DAM**
**SECTION AT SPILLWAY**

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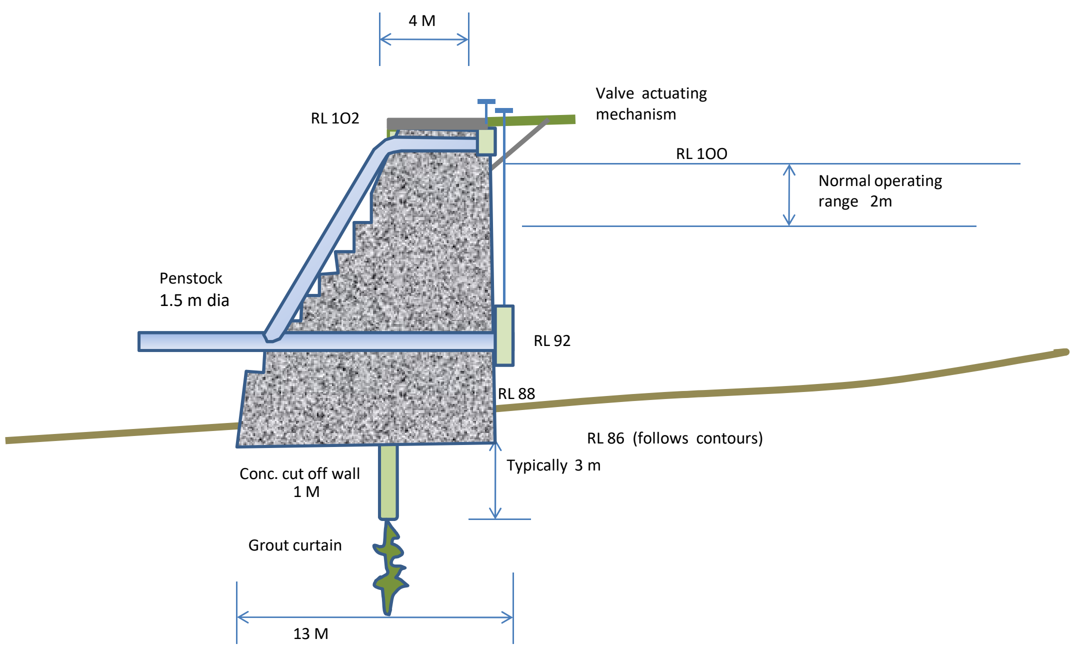


**ROLLED COMPACTED DAM**
**SECTION AT PENSTOCKS**

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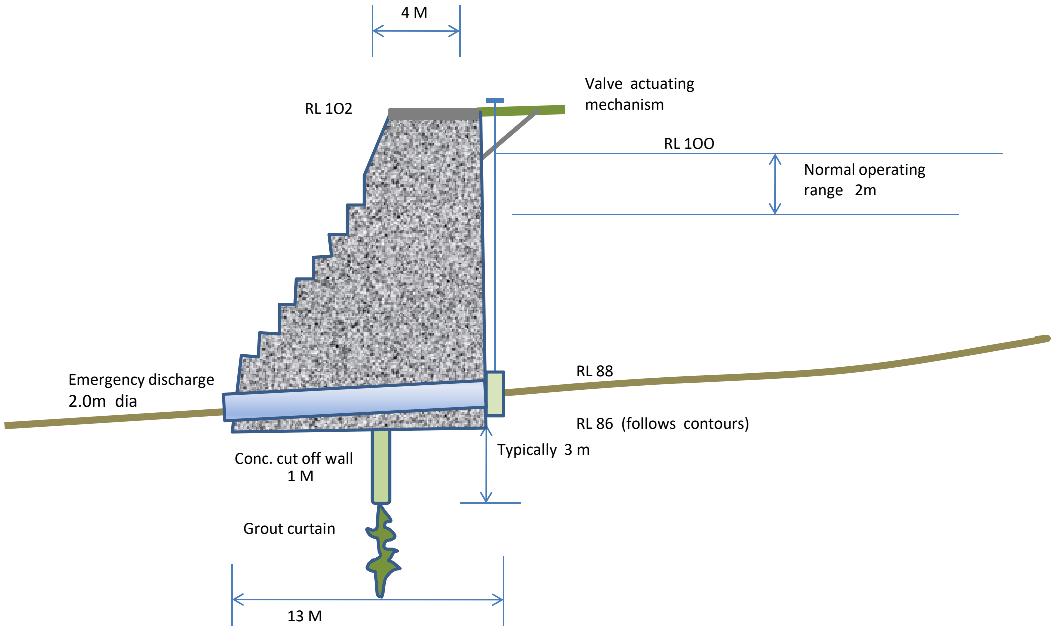


**ROLLED COMPACTED DAM**

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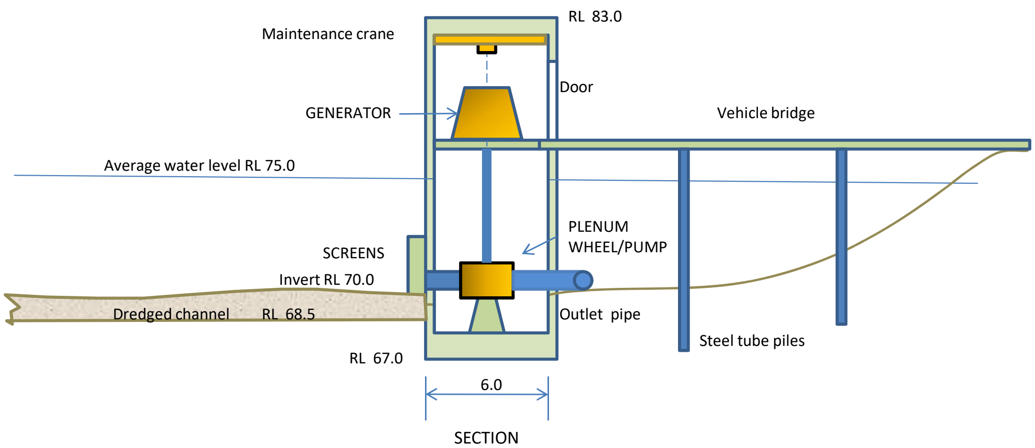


**PUMP STATION**

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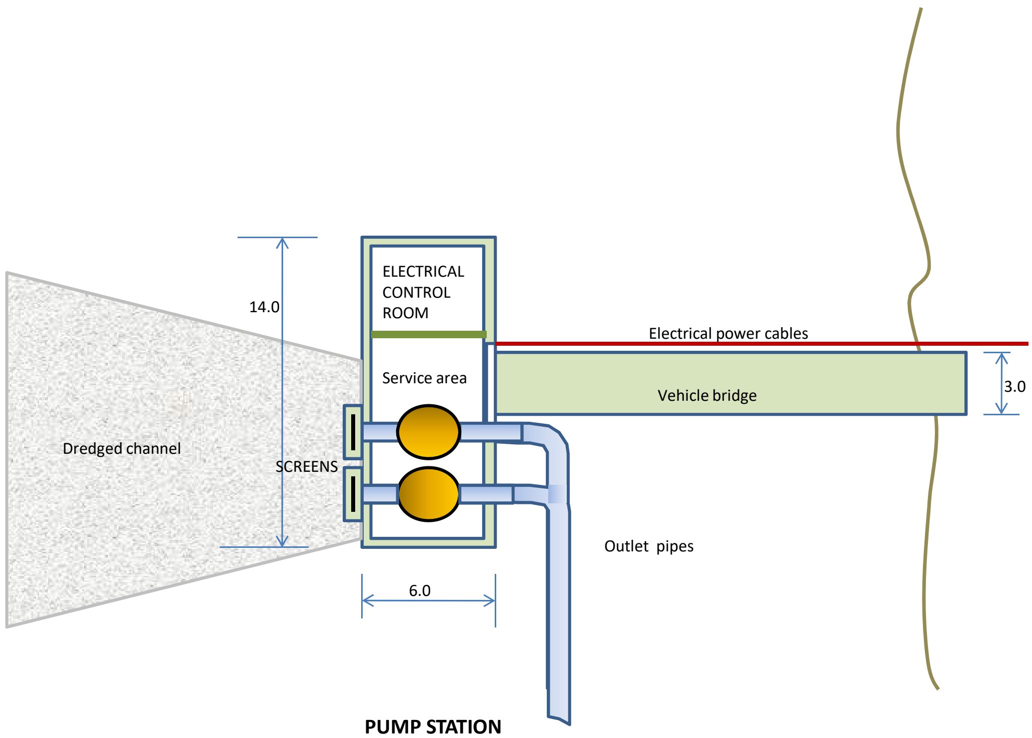


**PUMP STATION**

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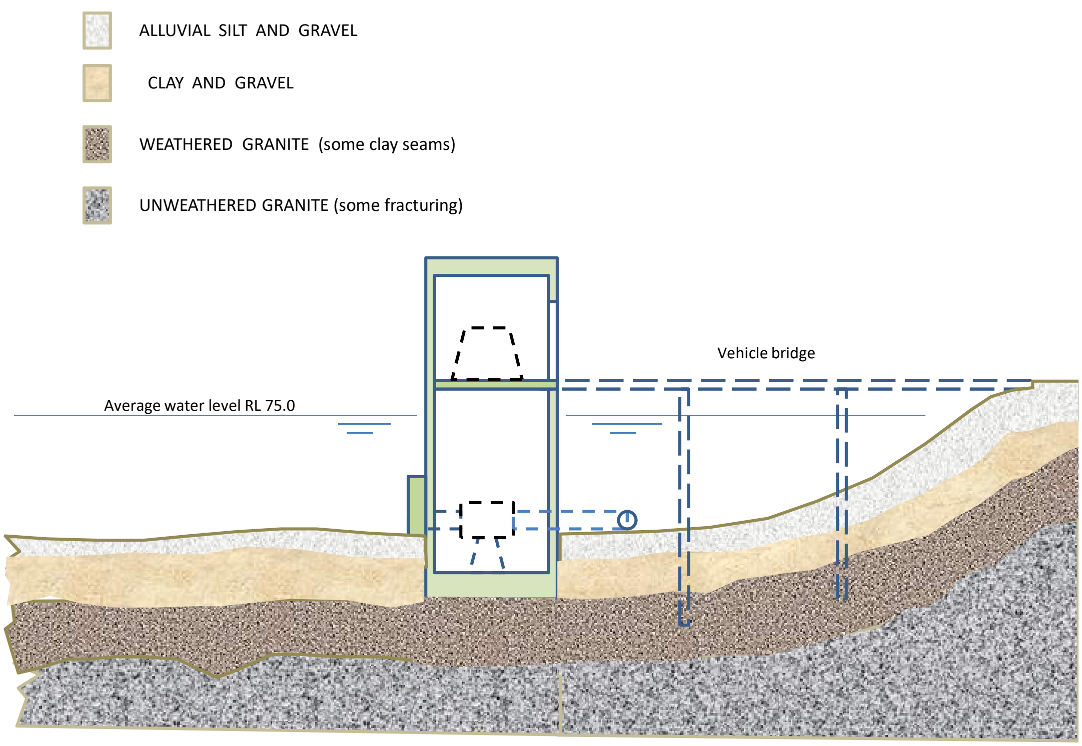


**SUB STRATA**

SECTION

Not to scale

**Note:**

These images are provided for conceptual understanding only.

**What you need to submit:**
**Submission requirements**
**You need to submit the report part 1 (group submission- Internal Briefing Document) and part 2 (individual submission- Reflective Report) electronically, in word format, via Canvas, by midnight on the due date (11th September at 23.59)**

## Criteria for Assessment

Refer to the CRA of group & individual assessment.

## Resources and Support Available

Four weeks of lectures, project briefing document, project data, and tutorial exercises are provided.

## Academic Integrity Statement

You are reminded to maintain academic integrity and professional standards in your submission in a manner which is fair, honest, and accountable. The use of artificial intelligence (GenAI) tools are prohibited for the assessment.